Kexin Wei

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Research Interests

- **Robotics & AI**: reinforcement learning for robot control and autonomy, medical and surgical robotics, and learning-based perception — building on hands-on experience with 4/6-DoF robot arms, motion tracking, and deep-learning image segmentation.
- **Supporting background**: medical imaging, cyber-physical systems security, and full-stack engineering for research platforms.

Education

M.Eng in Medical Robotics and AI — Aug 2020 — May 2022

National University of Singapore, Singapore — *GPA: 4.63/5.00*

- **Thesis**: Deep Reinforcement Learning (DRL) for Robot-assisted Surgical Training, which integrates DRL with a virtual laparoscopic surgery training system, wherein the DRL agent drives the surgical instrument and reaches goals of the surgical training tasks. Code.
- **Teaching Assistant**: ME2143E Motor Characteristics and ME5405 Machine Vision.

B.Eng in Mechanical Design, Manufacture and its Automation — Sep 2014 — Jul 2019

Tongji University, China — *GPA: 4.61/5.00*

- **Final Project**: ECU Control of Boiler Combustion System with HMI Interface — utilized a Freescale electric control unit receiving requests from a human interface, enabling users to monitor and control boiling procedures and components.

Professional Experience

Research Engineer — Mar 2025 — Present

Imperial Global Singapore (IGS) — IN-CYPHER Programme with NTU — *Singapore*

- **Security research**: Investigating agentic, LLM-driven automated penetration testing (Autopatch) for medical cyber-physical systems, with PACS/DICOM assessment on an Orthanc/OHIF testbed — contributing to a submitted platform paper.
- **Simulation**: Reproduced and validated a real-time closed-loop glucose-insulin simulator (OpenAPS/simglucose/UVA-Padova), translating MATLAB controllers to Python.
- **Research platform**: Built the evaluation platform end-to-end — Flask + Vue.js visualization, Dockerized multi-container deployment, and lab compute (Ubuntu/RHEL, KVM) supporting group experiments.

Software Team Lead — Mar 2024 — Jan 2025

Creative Medtech Solutions Pte.Ltd. — *Singapore*

- **Robot**: Engineered robotics control systems including kinematics optimization for 4-DoF robot and control interfaces.
- **CyberSecurity**: Enhanced security through encryption protocols, SQL injection protection, and streamlined Qt-based GUI.
- **Image**: Developed 4D DICOM processing and US/MRI registration algorithms for medical image alignment.
- **Lead**: Led a team of 6 engineers, driving Git workflow standardization, Scrum adoption, GoogleTest integration, and code review standards.

Research and Development Engineer — Jun 2022 — Mar 2024

Creative Medtech Solutions Pte.Ltd. — *Singapore*

- **Robot**: Designed respiratory motion tracking algorithm for 6-DoF robot arm integrating an A*STAR CNN algorithm, verified in vivo across pig and dog trials.
- **AI**: Developed and validated an AI-based 3D medical image segmentation pipeline for prostate biopsy guidance.
- **Image**: Prototyped kidney segmentation solutions using SAM, MedSAM, and OpenCV for potential implementation.
- **Signal Processing**: Created power calibration algorithm for High Intensity Focused Ultrasound treatment, verified in ex-vivo experiments.
- **Sensor**: Integrated electromagnetic sensor system in HIFU treatment software for 3D visualization of the ultrasound probe.
- **Patent**: Filed 3 patents in medical algorithm design: HIFU calibration, respiratory tracking, and time synchronization.

Research Assistant — Internship **Jul 2018 — Sep 2018**

Beijing Surgerii Technology Co., Ltd. and Kii Lab of SJTU under Professor Kai Xu *Shanghai, China*

- Conducted 20+ experiments to evaluate the three-dimensional positioning algorithm of a multi-backbone robot.
- Assembled and tested the motor-driven part of a surgical robot arm to ensure fundamental functions.
- Analyzed needle insertion experiment results and redesigned mechanical structure using AutoCAD for improved performance.

Publications & Patents

- **K. Wei**, I. Zahid, L. Cheng, A. Bird, V. Schlegel, and N. Oliver, "Generalisability of the Glucoprint: CGM-Based Re-identification Across Six Type 1 Diabetes Cohorts," Imperial College London / Imperial Global Singapore. *In preparation*.
- L. Cheng, **K. Wei**, V. Schlegel, Q. Shao, A. Bird, W. Wei, N. Oliver, and A.A. Bharath, "G-FM: Imputation-Guided Flow Matching for Multivariate Time-Series Generation," Imperial College London / Imperial Global Singapore. *Submitted*.
- Y. Niu, Y. Ma, S. Nandakumar, M. Chen, V. Schlegel, **K. Wei**, L. Cheng, A. Bird, A.A. Bharath, and S.K. Lam, "HealthLoopQA: A Context-Aware Question Answering Benchmark for Interpreting Wearable Monitoring Data in Diabetes Care," Imperial Global Singapore / Imperial College London / NTU. *Submitted*.
- A.R. Bird*, V. Mohan*, **K. Wei**, Z. Cao, S.H. Ng, M.M. Mulay, V. Schlegel, M.G. Moffett, J.E. Moore Jr, C.-H. Chang, N. Oliver, and A.A. Bharath, "Toward Trustworthy Closed-Loop Diabetes Technologies: A Dynamic Platform for Safety and Security Evaluation," Imperial Global Singapore / Imperial College London / NTU. *Submitted*.
- **3 Patents Filed**: HIFU power calibration algorithm, robot respiratory tracking algorithm, and HIFU sweep mode time synchronization.

Awards & Honors

AI	Hackathon "Runner Up" Prize at bio LLMs hosted by 4Catalyzer (2023)
Mathematics Modeling	2nd Prize in CUMCM (2018); Honorable Mention in MCM/ICM (2018)
Scholarship	3rd Class Award (2018); 1st Class Award (2016); 3rd Class Award (2015)

Skills

Programming	Python, C++, Matlab, SQL, Rust
Frameworks	Qt, Flask, Vue.js, Flutter, OpenCV, VTK
AI/ML	PyTorch, TensorFlow, Claude Agent SDK, LangChain/LangGraph, OpenAI & Gemini APIs
Tools & Infra	Docker, Git, Linux (Ubuntu/RHEL), ROS, CI/CD
Language	English (C1), German (C1), Mandarin (native)

Community & Service

- **Open source**: Contributor to Women Devs SG projects — the community website (merged) and the *bibsnbub* childcare-finder app.
- **Junior Developers SG**: Participant in coding dojos — test-driven development, refactoring, and test automation.
- **Division Zero (div0)**: Cybersecurity workshops — home-lab building, SHELLgym, and Hack Your First Flag.